

Sky130 ASIC Characterization and Yield Prediction Project

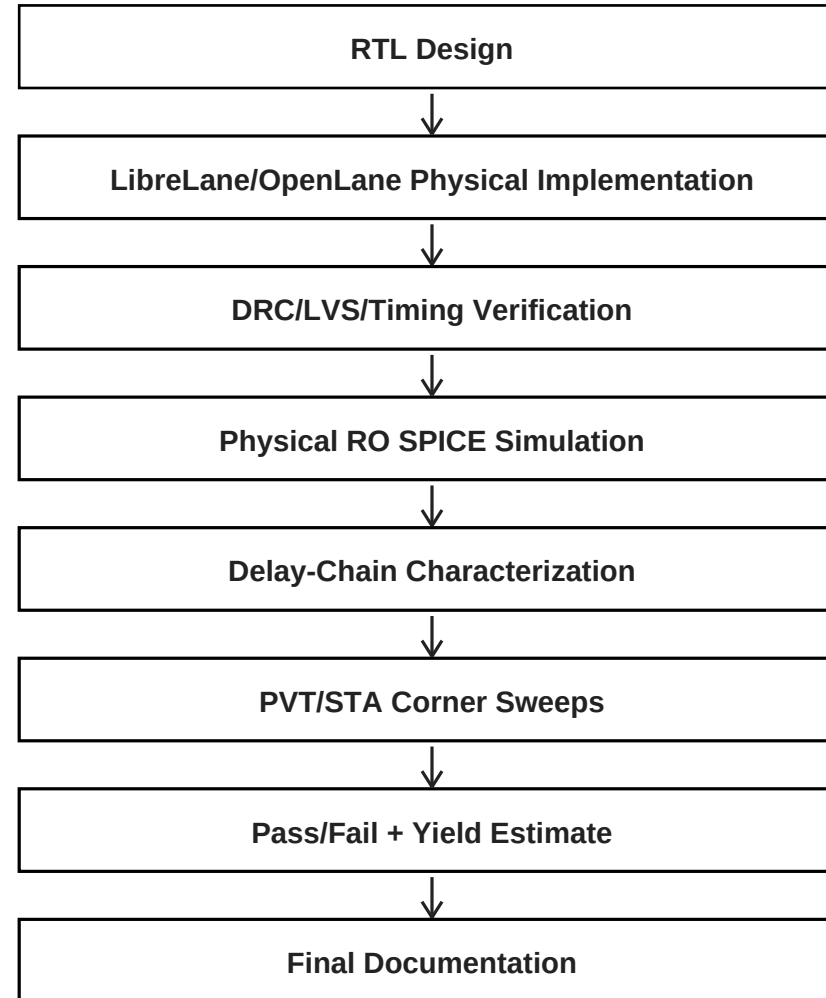
Clean final report with physical RO SPICE, post-layout extraction, delay-chain verification, PVT sweeps, and estimated yield

Praneeth Satishkumar

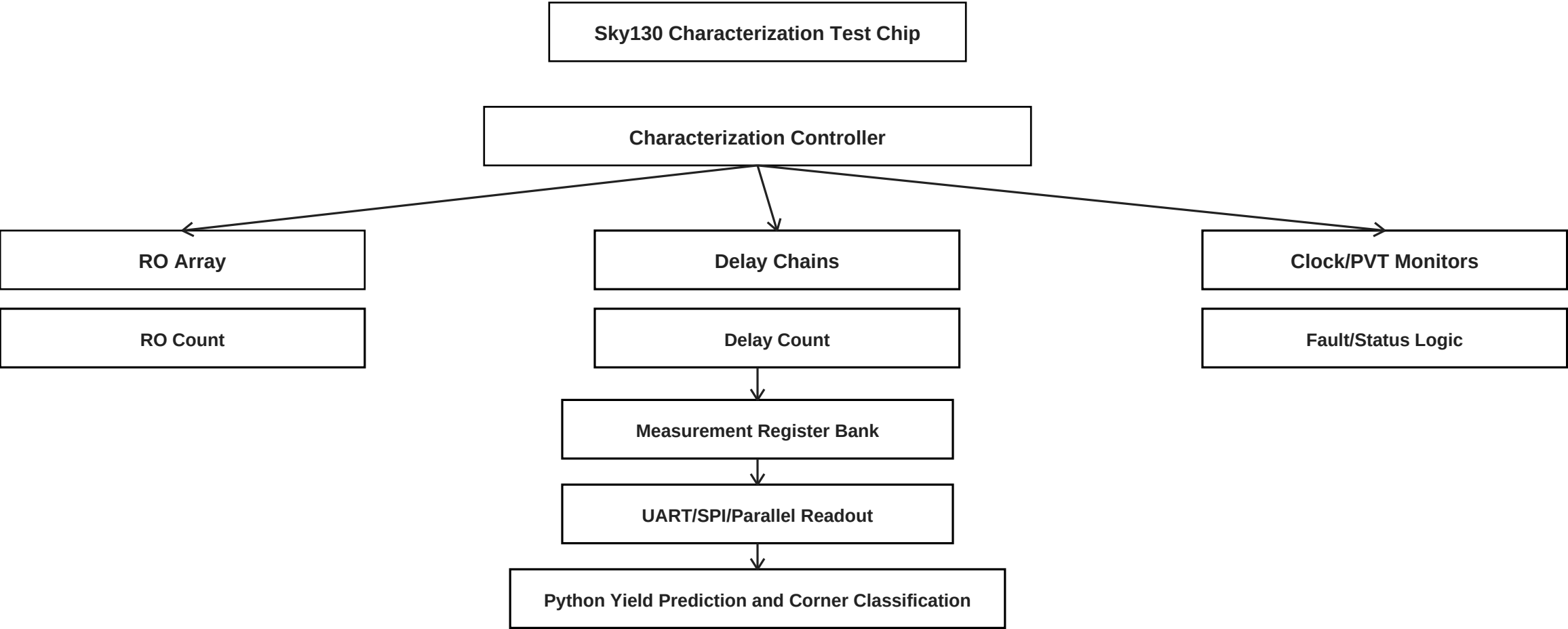
Project Summary

- Designed a Sky130 characterization test chip flow around ring oscillator behavior, delay-chain timing, and STA corner checks.
- Implemented the main RTL-to-layout flow using LibreLane/OpenLane with Sky130 standard cells.
- Completed PVT/STA sweeps for available process, voltage, and temperature libraries.
- Replaced old simulated RO input-clock data with physical Sky130 RO SPICE data.
- Completed TT post-layout extracted RO simulation and TT post-layout extracted delay-chain simulation.
- Generated simulation/STA-based yield estimates; this is not measured silicon yield.

Final Project Workflow



Characterization Test Chip Architecture



Physical Implementation and Verification Summary

Check	Result	Status
Sky130 RTL-to-layout flow	Completed using LibreLane/OpenLane	Pass
Final GDS/layout	Generated and opened in KLayout	Pass
Magic DRC	0 errors for verified blocks	Pass
LVS	Circuits match uniquely	Pass
Post-route STA	0 setup violations, 0 hold violations in baseline flow	Pass
Lint	Warnings documented; no must-fix issue blocking final results	Documented

The main characterization design and the physical RO/delay-chain blocks were implemented and verified using the Sky130 flow. The RO and delay-chain physical blocks were also used for SPICE-level evidence.

Sky130 Corner Timing Checks

Corner	Voltage	Temperature	Setup Slack	Hold Slack	TNS	WNS	Status
TT	1.80 V	25/27 C	6.17 ns	0.23 ns	0.00	0.00	Pass
SS	1.60 V	100 C	4.02 ns	0.72 ns	0.00	0.00	Pass
FF	1.95 V	-40 C	6.80 ns	0.05 ns	0.00	0.00	Pass

FS and SF process corners were searched for but were not available in the installed Sky130 standard-cell Liberty files. The process sweep therefore uses the available TT, SS, and FF timing corners.

Voltage Sweep - FF at -40 C

Process	Temperature	Voltage	Liberty File	Setup Slack	Hold Slack	WNS	TNS	Status
FF	-40 C	1.56 V	ff_n40C_1v56.lib	6.18	0.19	0	0	Pass
FF	-40 C	1.65 V	ff_n40C_1v65.lib	6.41	0.15	0	0	Pass
FF	-40 C	1.76 V	ff_n40C_1v76.lib	6.59	0.10	0	0	Pass
FF	-40 C	1.95 V	ff_n40C_1v95.lib	6.80	0.05	0	0	Pass

The FF voltage sweep passed at all tested voltages. Setup slack increased with voltage, while hold slack became tighter as the fast corner became more aggressive.

Voltage Sweep - SS at -40 C

Process	Temperature	Voltage	Liberty File	Setup Slack	Hold Slack	WNS	TNS	Status
SS	-40 C	1.28 V	ss_n40C_1v28.lib	-13.24	-1.42	-13.24	-1235.32	Fail
SS	-40 C	1.35 V	ss_n40C_1v35.lib	-5.62	-0.49	-5.62	-214.25	Fail
SS	-40 C	1.40 V	ss_n40C_1v40.lib	-2.39	-0.08	-2.39	-44.41	Fail
SS	-40 C	1.44 V	ss_n40C_1v44.lib	-0.57	0.16	-0.57	-4.74	Fail
SS	-40 C	1.60 V	ss_n40C_1v60.lib	3.27	0.72	0	0	Pass
SS	-40 C	1.76 V	ss_n40C_1v76.lib	5.15	0.47	0	0	Pass

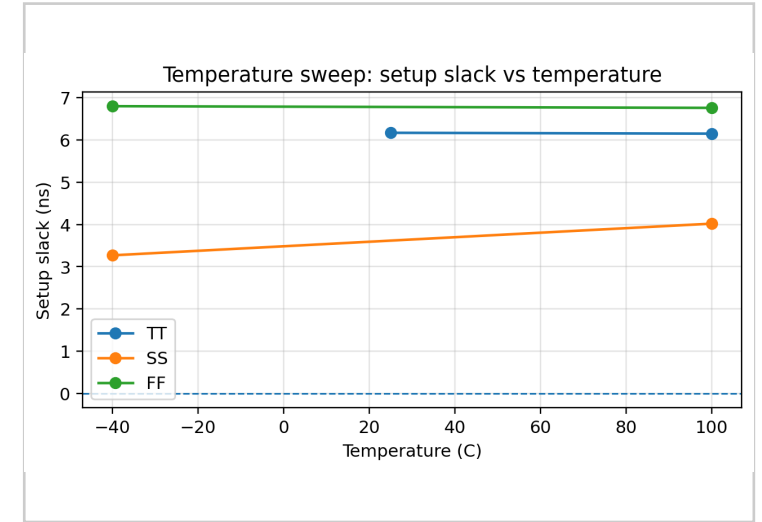
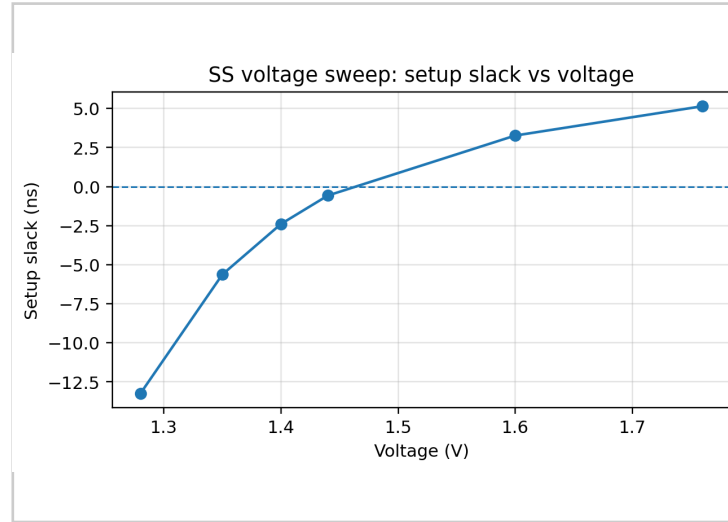
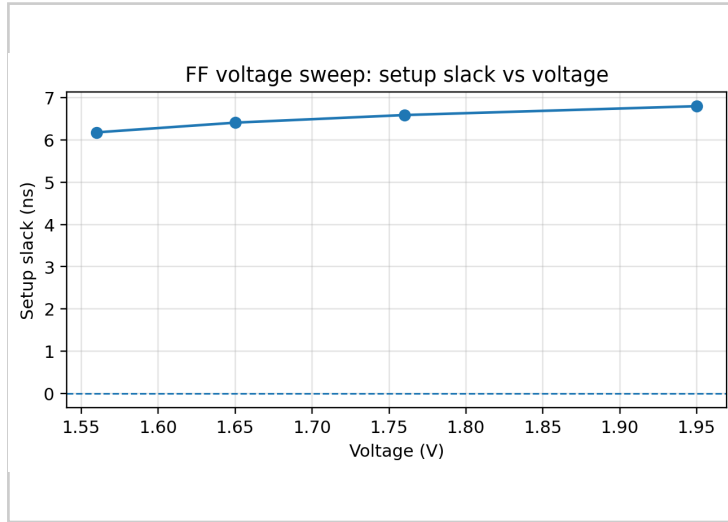
Low-voltage SS cases below 1.60 V failed setup timing. This creates the main failure group used in the estimated yield table.

Temperature Sweeps

Process	Voltage	Temperature	Liberty File	Setup Slack	Hold Slack	WNS	TNS	Status
TT	1.80 V	25 C	tt_025C_1v80.lib	6.17	0.23	0	0	Pass
TT	1.80 V	100 C	tt_100C_1v80.lib	6.15	0.24	0	0	Pass
SS	1.60 V	-40 C	ss_n40C_1v60.lib	3.27	0.72	0	0	Pass
SS	1.60 V	100 C	ss_100C_1v60.lib	4.02	0.72	0	0	Pass
FF	1.95 V	-40 C	ff_n40C_1v95.lib	6.80	0.05	0	0	Pass
FF	1.95 V	100 C	ff_100C_1v95.lib	6.76	0.07	0	0	Pass

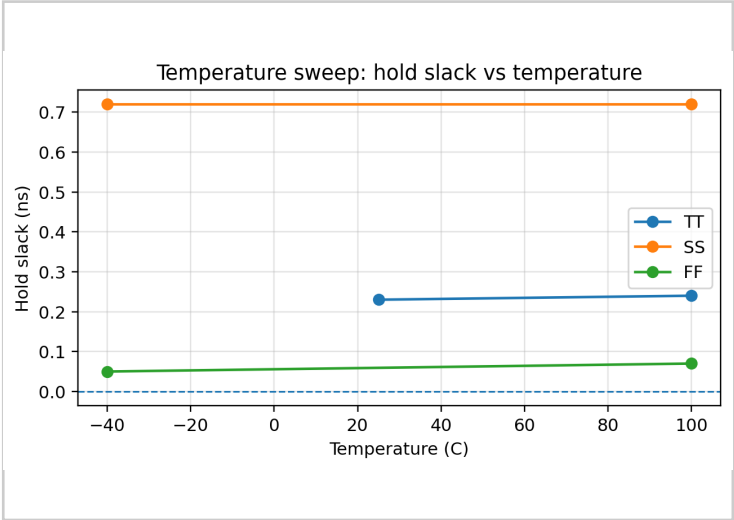
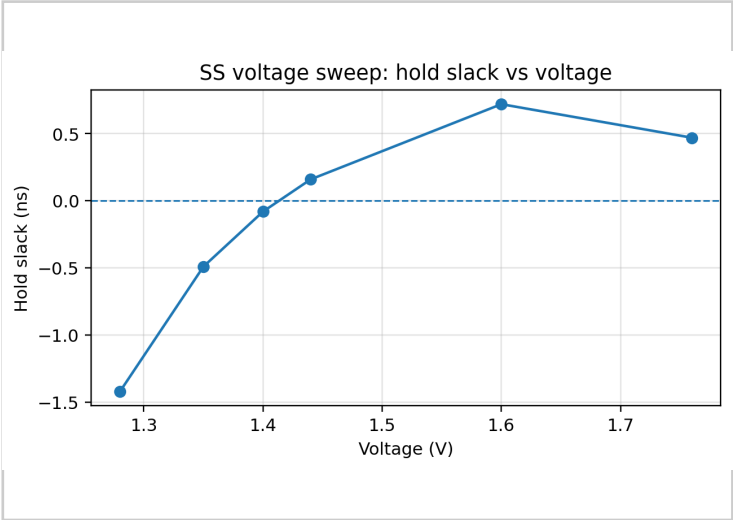
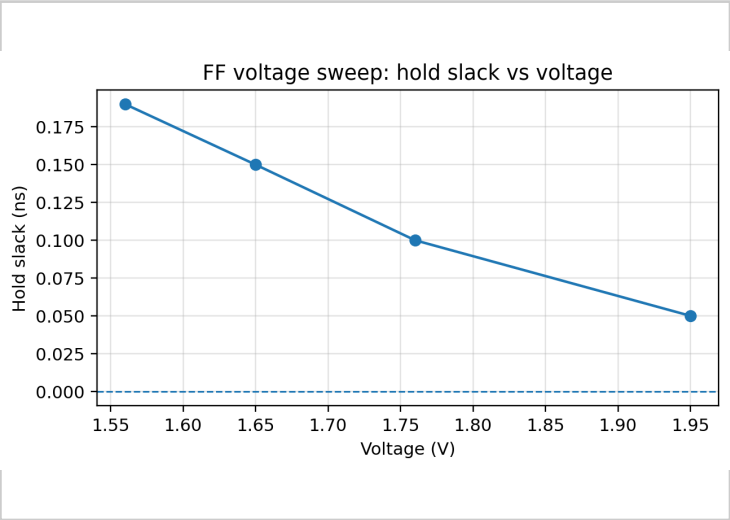
Temperature sweeps passed for all available matched temperature pairs. FF remained the most hold-sensitive case.

Plots - Voltage/Temperature vs Setup Slack



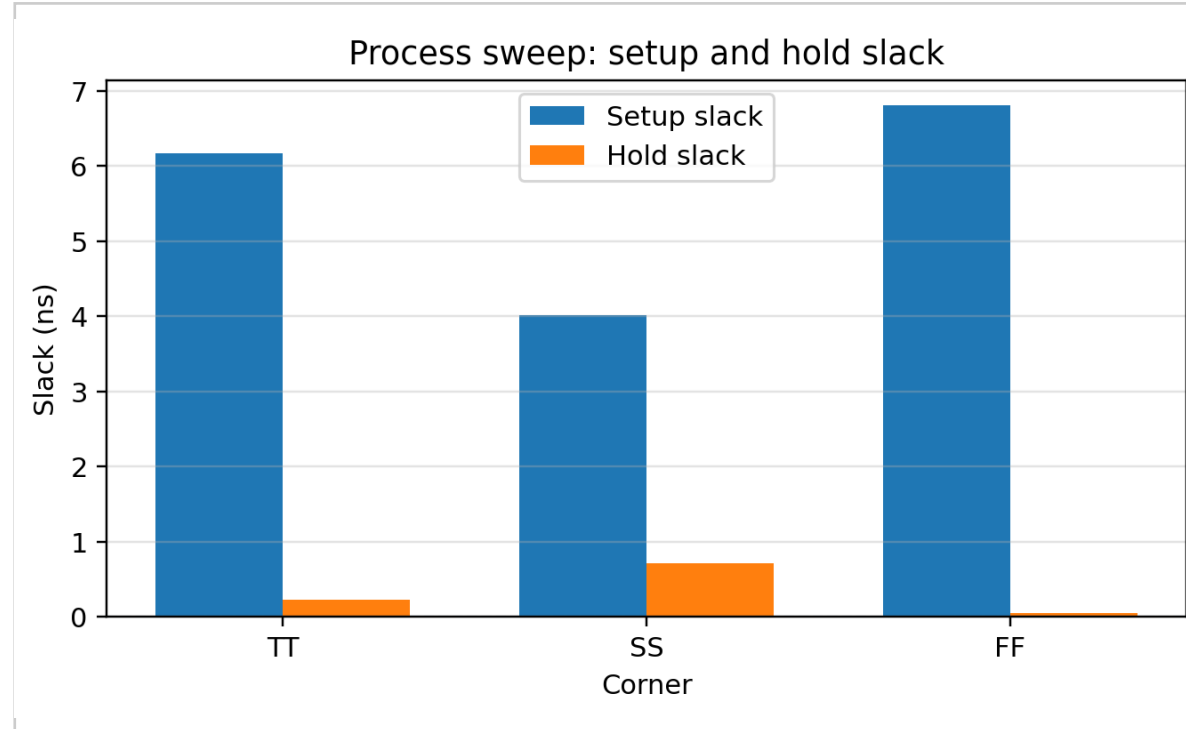
Setup slack improves as SS voltage increases. The SS low-voltage cases define the main timing failure boundary. Temperature sweeps remained timing-clean for the available paired libraries.

Plots - Voltage/Temperature vs Hold Slack



FF has the tightest hold behavior, with hold slack reaching 0.05 ns. This matches the expected fast-corner behavior where hold timing can become more sensitive.

Process Sweep Plot



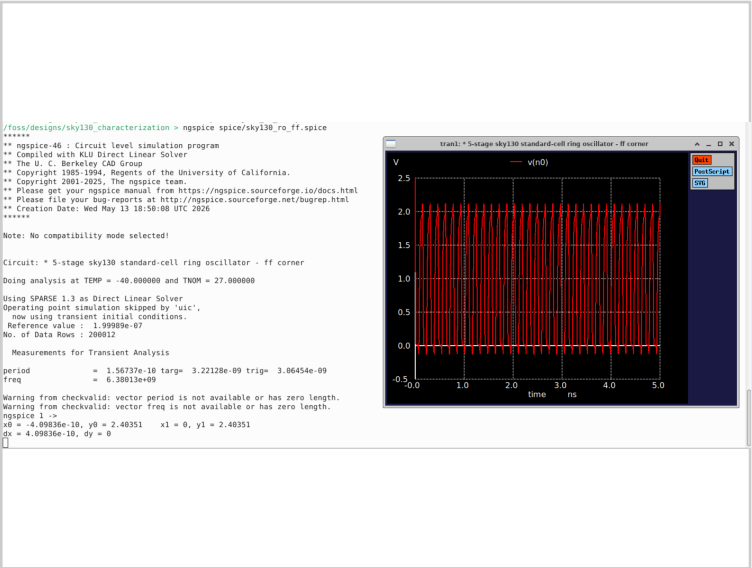
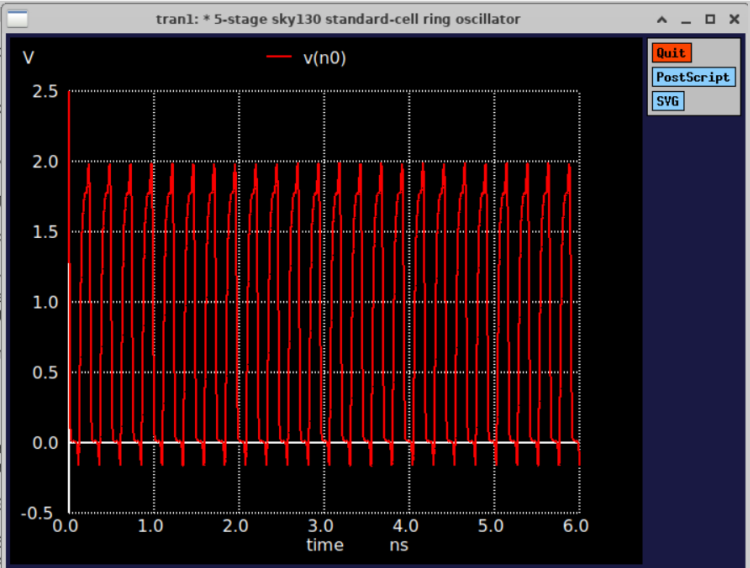
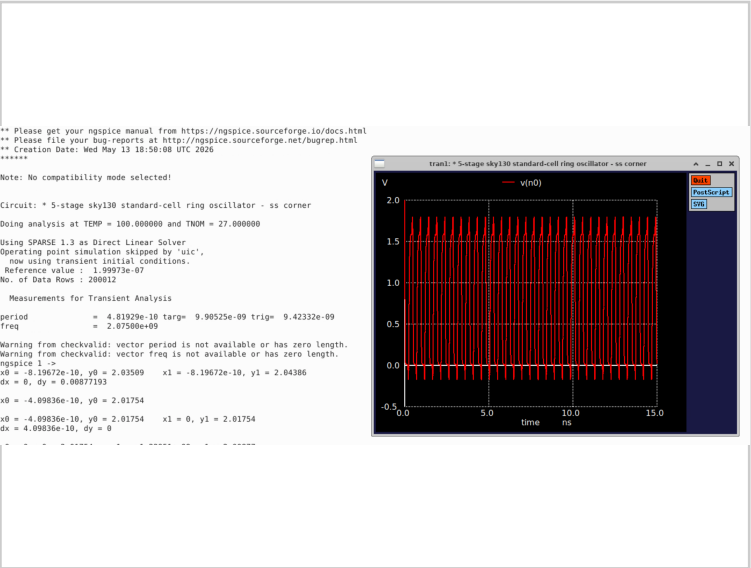
SS produced the lowest setup slack among TT/SS/FF and FF produced the tightest hold slack. This supports the slow/typical/fast classification.

Final RO Frequency Table

Corner	Voltage	Temperature	RO Type	Period	Frequency	Classification
SS	1.60 V	100 C	Pre-layout Sky130 SPICE	481.9 ps	2.075 GHz	Slow
TT	1.80 V	27 C	Pre-layout Sky130 SPICE	245.5 ps	4.073 GHz	Typical
TT	1.80 V	27 C	Post-layout extracted SPICE	505.69 ps	1.9768 GHz	Typical / extracted
FF	1.95 V	-40 C	Pre-layout Sky130 SPICE	156.7 ps	6.380 GHz	Fast

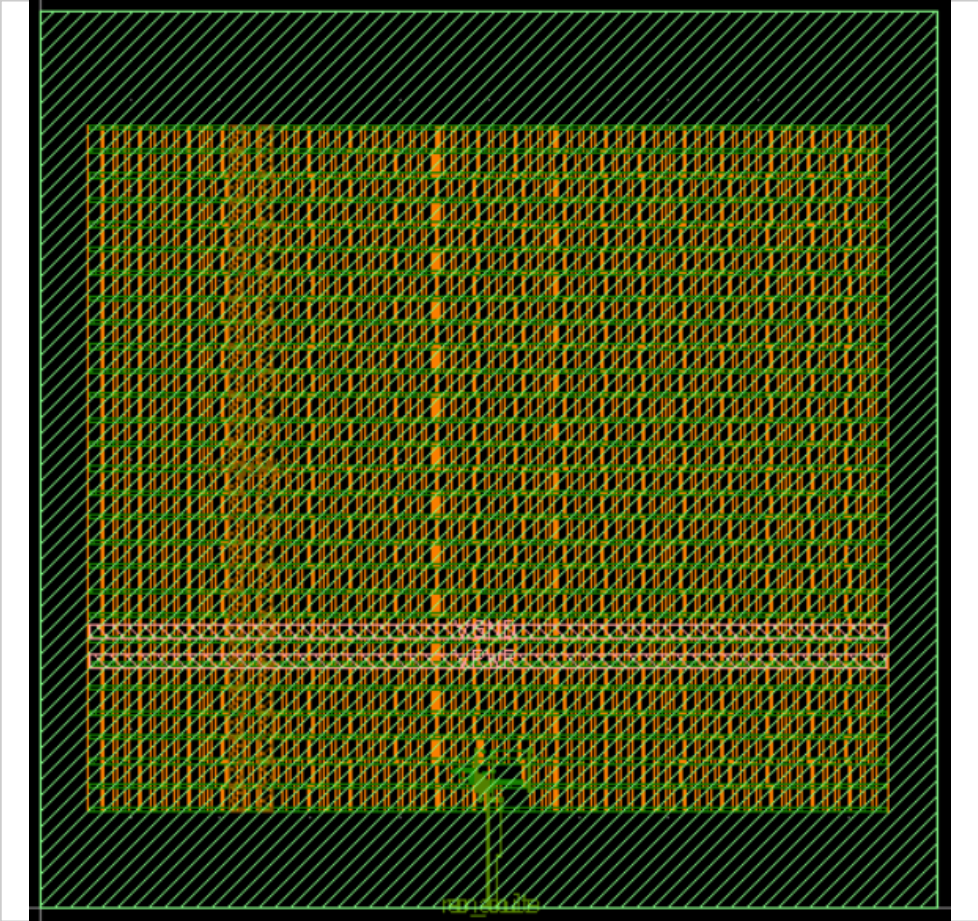
The old simulated RO input-clock values (80/150/230 counts and 8/15/23 MHz) were removed from the final characterization and retained only as counter functional verification. Final RO characterization uses Sky130 SPICE data.

Physical RO SPICE Waveforms



Sky130 5-stage standard-cell ring oscillator waveforms confirm oscillation across SS, TT, and FF. SS produces the lowest frequency, TT is nominal, and FF produces the highest frequency.

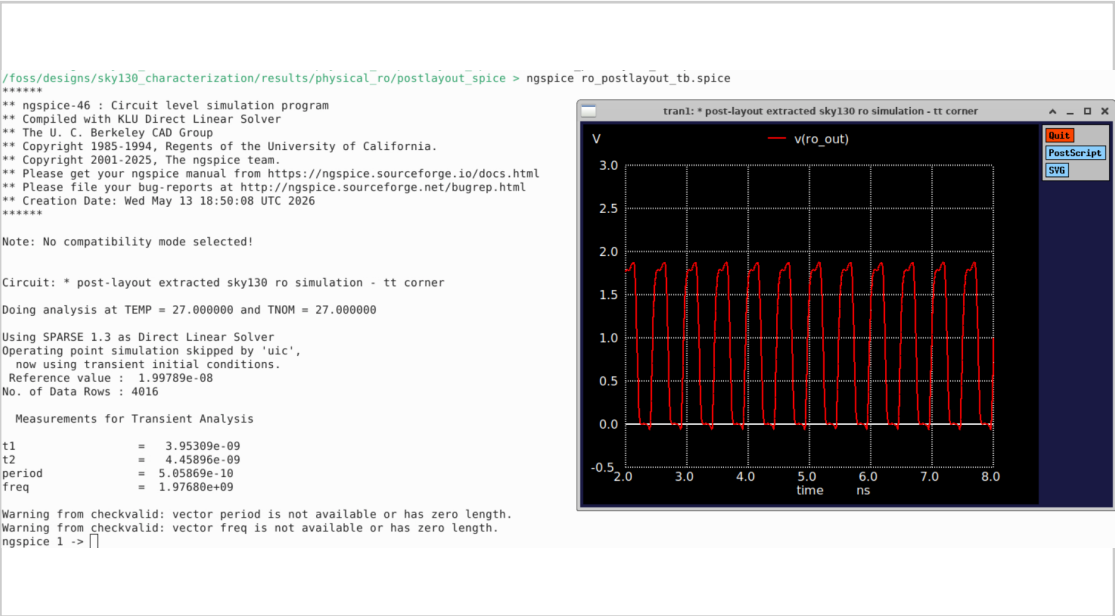
Physical Verification - Ring Oscillator



A real Sky130 standard-cell ring oscillator was physically implemented through LibreLane. The RO-only layout passed Magic DRC with 0 errors and LVS with circuits matching uniquely. This proves physical implementation, while frequency data comes from Sky130 SPICE and TT post-layout extracted SPICE simulation.

Check	Result
Final RO GDS	Generated
Magic DRC	0 errors
LVS	Circuits match uniquely
TT extracted SPICE	Completed

Post-layout Extracted RO Simulation - TT

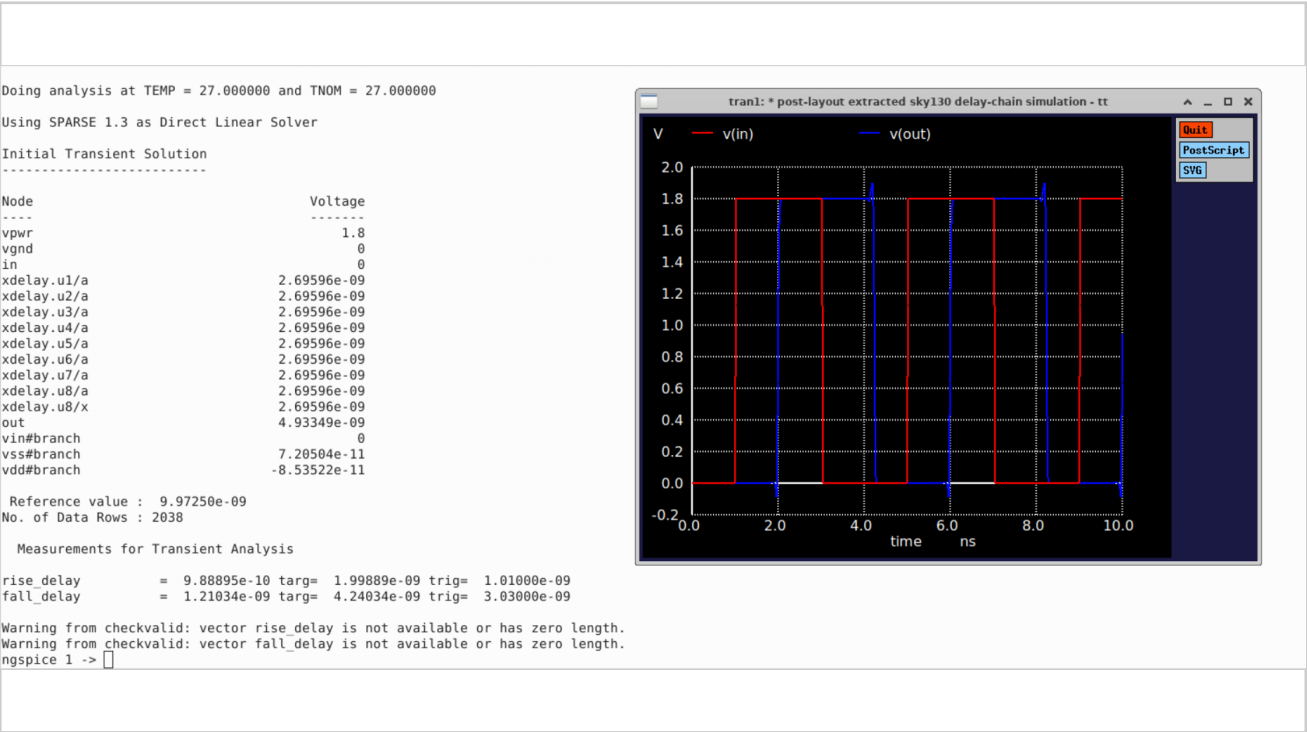
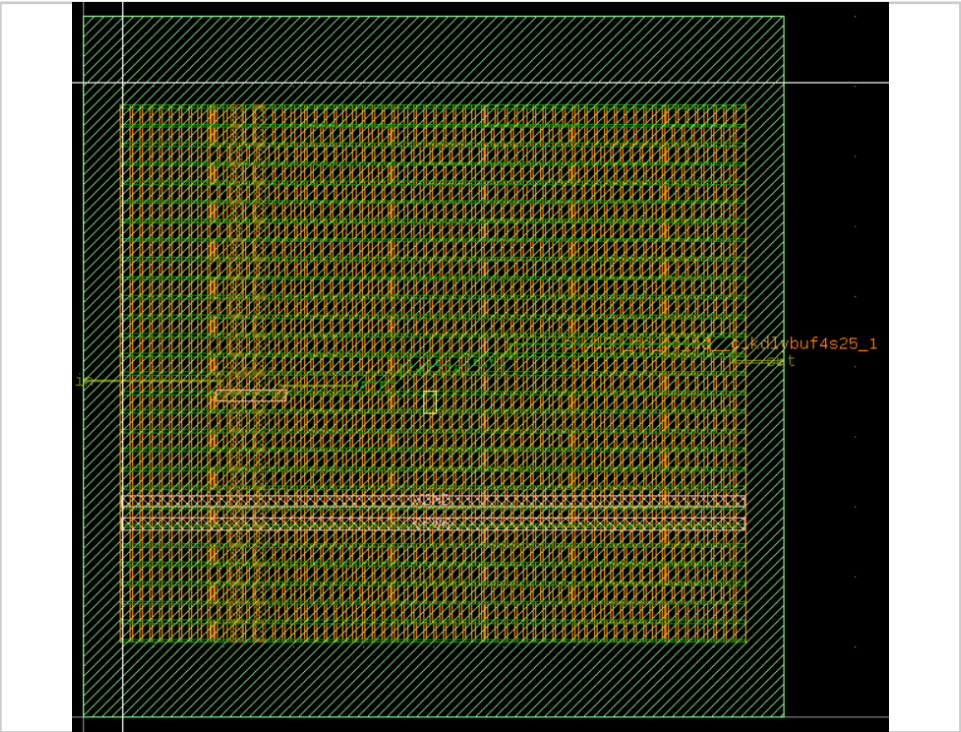


```
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14:.subckt sky130_fd_sc_hd_nand2_1 VPB VNB VGND VPWR A Y B a_113_47#
45:.subckt sky130_fd_sc_hd_ckdlybuf4s25_1 VNB VPB VPWR VGND A X a_244_47# a_355_47#
91:.subckt sky130_fd_sc_hd_inv_1 VPB VNB VGND VPWR A Y
111:.subckt sky130_fd_sc_hd_buf_2 VPWR VGND X A VPB VNB a_27_47#
141:.subckt ro_physical_top VGND VPWR enable ro_out
```

RO Simulation Type	Corner	Voltage	Temperature	Period	Frequency	Notes
Pre-layout SPICE RO	TT	1.80 V	27 C	245.5 ps	4.073 GHz	Standard-cell SPICE only
Post-layout extracted RO	TT	1.80 V	27 C	505.69 ps	1.9768 GHz	Extracted layout parasitics included

Post-layout extracted RO simulation was completed for TT. SS/FF extracted RO simulation and true silicon-measured RO data remain future work.

Physical Verification - Delay Chain



Corner	Voltage	Temperature	Rise Delay	Fall Delay	DRC	LVS	Status
TT	1.80 V	27 C	988.95 ps	1.21034 ns	0 errors	Match	Done

SS/FF delay-chain behavior is marked as expected/STA-based because extracted physical delay was completed for TT only.

Final Characterization Table

Corner	Voltage	Temperature	Physical RO Frequency	Delay Result	Setup Slack	Hold Slack	PVT Status	Classification	Pass/Fail
SS	1.60 V	100 C	2.075 GHz	Slow behavior expected / STA-based	4.02 ns	0.72 ns	Slow	Slow	Pass
TT	1.80 V	27 C	1.9768 GHz post-layout extracted	Rise 988.95 ps, Fall 1.21034 ns	6.17 ns	0.23 ns	Normal	Typical	Pass
FF	1.95 V	-40 C	6.380 GHz	Fast behavior expected / STA-based	6.80 ns	0.05 ns	Fast	Fast	Pass

This table combines physical RO frequency, delay-chain timing, STA timing result, PVT status, classification, and pass/fail result.

Yield Criteria

Check	Pass Criteria	Fail Criteria
Setup slack	Setup slack > 0 ns	Setup slack < 0 ns
Hold slack	Hold slack > 0 ns	Hold slack < 0 ns
WNS	WNS >= 0 ns	WNS < 0 ns
TNS	TNS = 0	TNS < 0
DRC	0 violations	More than 0 violations
LVS	Circuits match uniquely	LVS mismatch
Physical RO SPICE	Oscillation observed and frequency measured	No oscillation or invalid measurement
Delay-chain SPICE	Rise/fall delays measured and within expected operation	Invalid or missing delay measurement

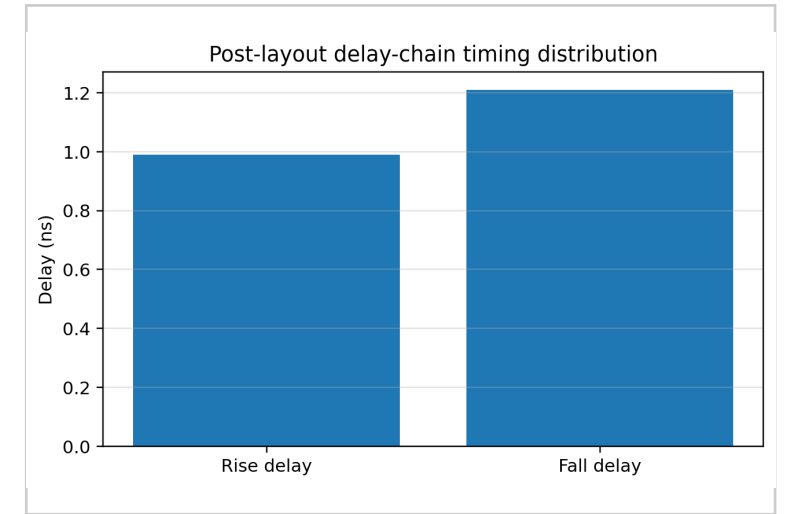
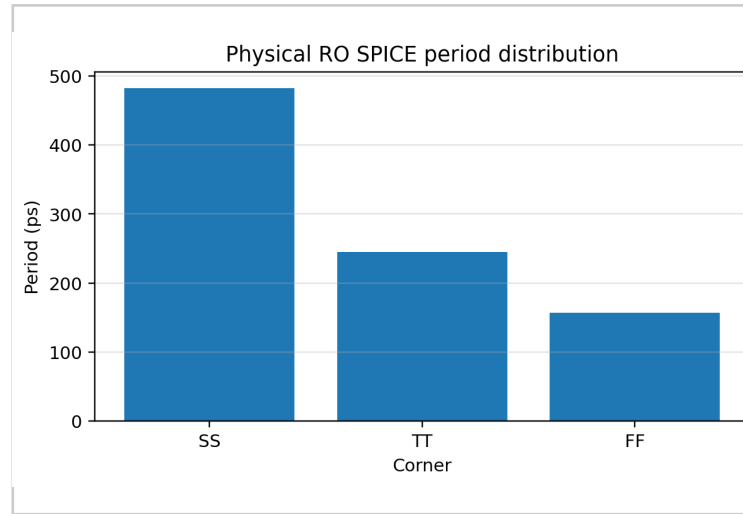
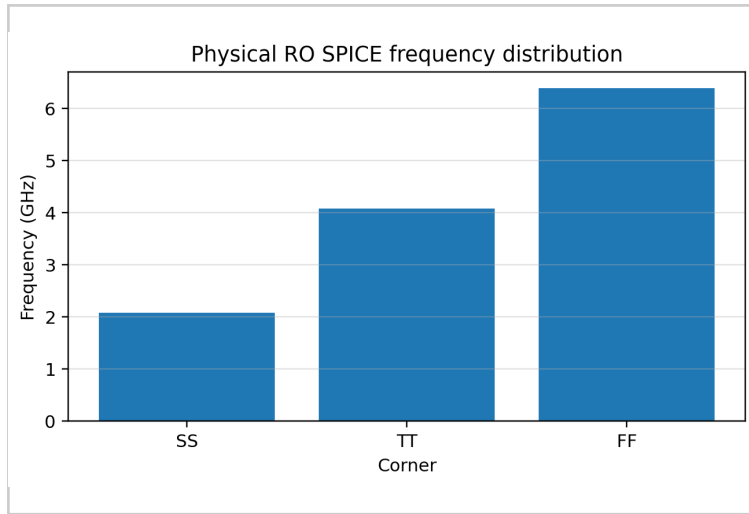
These criteria are used to compute a simulation/STA-based estimated pass rate. They do not represent measured silicon yield.

Updated Yield Prediction Table

Category	Passing Cases	Total Cases	Estimated Pass Rate
Overall STA sweep	15	19	78.9%
Physical RO SPICE sweep	3	3	100.0%
Post-layout extracted RO	1	1	100.0%
Physical delay-chain SPICE	2	2	100.0%
Physical verification	4	4	100.0%

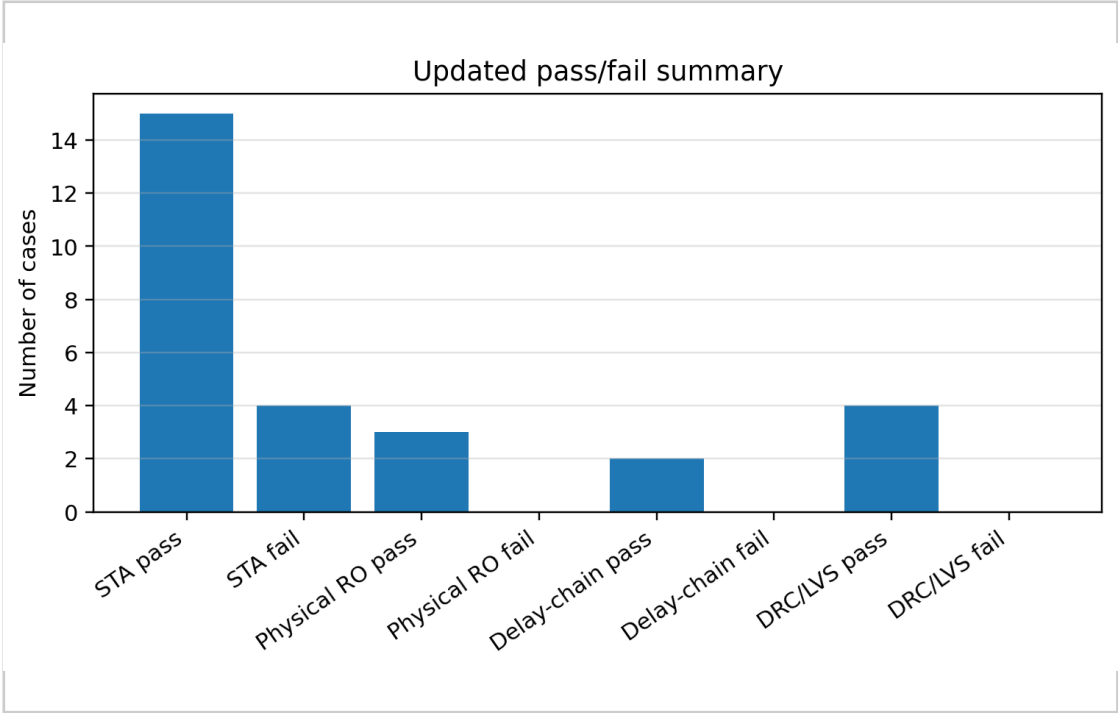
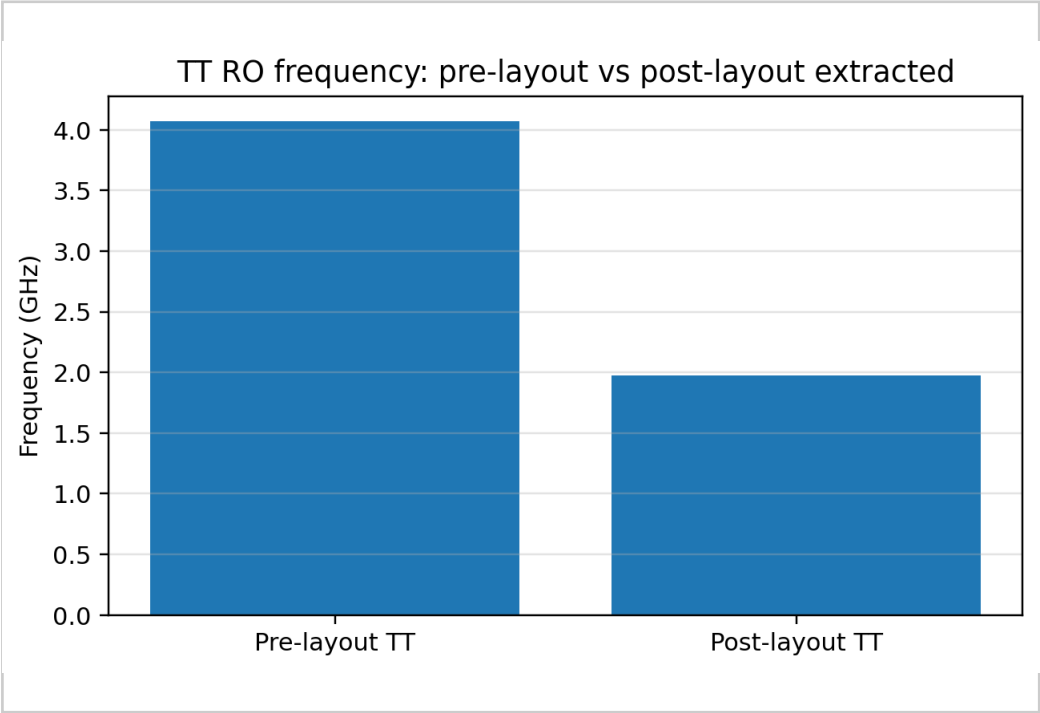
Label: Simulation/STA-based estimated yield, not silicon yield. The only failures in the updated yield table come from SS low-voltage STA cases.

Updated Plots - RO and Delay



The old RO plot based on simulated input clocks was replaced by the physical RO SPICE frequency plot. The delay-chain plot uses the TT post-layout extracted rise/fall delay values.

Updated Plots - Post-layout Comparison and Pass/Fail



The TT post-layout extracted RO is slower than the pre-layout TT RO because extracted layout parasitics are included. The pass/fail plot includes SS low-voltage STA failures and confirms no physical RO or delay-chain failures in the completed simulations.

Slow / Typical / Fast Classification

Behavior	Evidence Used	Result
Slow	SS corner, lowest RO frequency, low-voltage/high-temperature stress, lower setup slack	SS classified as slow
Typical	TT corner, nominal voltage, clean timing, stable RO oscillation	TT classified as typical
Fast	FF corner, highest RO frequency, high setup slack, tightest hold slack	FF classified as fast

Rules: Low RO frequency + worse timing behavior = slow. Nominal RO frequency + clean timing = typical. High RO frequency + high setup slack and tighter hold slack = fast.

Comparison with Expected Sky130 Behavior

Expected Behavior	Observed Result	Match
SS / low-voltage corners should be slower	SS RO frequency was lowest at 2.075 GHz; SS low-voltage STA cases failed below 1.60 V	Yes
TT should be nominal baseline	TT passed STA timing and produced stable RO oscillation	Yes
FF should be fastest	FF RO frequency was highest at 6.380 GHz	Yes
Higher voltage should improve setup slack	SS setup slack improved from failures at 1.28-1.44 V to passes at 1.60 V and 1.76 V	Yes
Hold slack can become tighter in fast corners	FF hold slack was lowest at 0.05 ns	Yes

Final Limitations

- Results are based on Sky130 STA, layout, and SPICE simulation.
- True silicon-measured RO data requires fabrication, tapeout, and measurement from actual silicon.
- Post-layout extracted RO simulation was completed for TT only.
- SS/FF extracted RO simulations remain future work and can improve corner accuracy.
- Delay-chain post-layout SPICE simulation was completed for TT only; SS/FF delay behavior is expected/STA-based.
- This is not full Monte Carlo silicon yield; it is a corner-based simulation/STA estimated yield.

Final Deliverables Completed

Deliverable	Status
RTL design and architecture	Done
Sky130 physical implementation	Done
DRC/LVS/timing verification	Done
PVT/STA process, voltage, and temperature sweeps	Done
Physical RO SPICE sweep SS/TT/FF	Done
TT post-layout extracted RO simulation	Done
Physical delay-chain layout and TT post-layout SPICE delay	Done
Final characterization table	Done
Simulation/STA-based yield prediction	Done
Clean plots and documentation	Done

Final Result

The project now contains a corrected and complete Sky130 characterization flow with physical RO SPICE results, TT post-layout extracted RO simulation, physical delay-chain verification, PVT/STA sweeps, slow/typical/fast classification, and simulation/STA-based estimated yield.

STA sweep

15/19 pass

78.9%

Physical RO SPICE

3/3 pass

100%

TT extracted RO

1.9768 GHz

post-layout

Delay chain

988.95 ps / 1.21034 ns

rise/fall